

Lab 0: Development Environment Setup — Windows

Standalone share copy — self-contained; no course repository links required.

OS: Windows 10 / 11

Primary IDE: IntelliJ IDEA Community Edition

Optional IDE: VS Code (if you already prefer it)

Install tools on **your Windows laptop**. Shared cloud services are not needed until Week 4+.

What you install

Tool	Target
IntelliJ IDEA Community	Latest stable (primary)
VS Code (optional)	Latest stable + Extension Pack for Java
Temurin OpenJDK	21 LTS
Maven	3.9.x
Git	2.x

Verified reference layout: Temurin 21 at `C:\Program Files\Eclipse Adoptium\jdk-21`; Maven at `C:\Program Files\Apache\maven\current`; workspace `%USERPROFILE%\java-bootcamp`.

Steps

Step 1 — Install IntelliJ IDEA Community (primary)

1. Download **Community Edition** for Windows: <https://www.jetbrains.com/idea/download/>
2. Run the installer. Prefer options that add a desktop/Start menu shortcut.
3. Launch IntelliJ once and finish the first-run wizard.

Expected: Welcome screen opens.

Step 2 — Optional: Install VS Code

Only if you already prefer VS Code:

1. Install from <https://code.visualstudio.com/>
2. Extensions: **Extension Pack for Java** (`vscjava.vscode-java-pack`).

Step 3 — Install Temurin JDK 21 and set `JAVA_HOME`

1. Download Temurin 21 **JDK** Windows x64 **MSI**: <https://adoptium.net/temurin/releases/?version=21&os=windows&arch=x64&package=jdk>
2. Run the MSI as Administrator. Typical path: `C:\Program Files\Eclipse Adoptium\jdk-21`.
3. Set **system** environment variables (Win + R → `sysdm.cpl` → Advanced → Environment Variables):

- `JAVA_HOME` = that JDK folder
 - Prefixed Path entry: `%JAVA_HOME%\bin`
4. Remove other JDKs from Path if present.
 5. Open a **new** PowerShell window.

```
java -version
javac -version
echo $env:JAVA_HOME
```

Expected: Versions show `21.x.x` (Temurin / Eclipse Adoptium). Verified on authoring PC: Temurin `21.0.11`, `JAVA_HOME=C:\Program Files\Eclipse Adoptium\jdk-21`.

If it fails — “This environment variable is too large” (2047 chars): the Windows GUI cannot edit a Path longer than 2047 characters. Click **Cancel** (do not OK). Prefer PowerShell as Administrator to trim duplicates / user-folder entries from the **System** Path, or move rarely used tools to the **User** Path. Do not use `setx` for Path (it truncates). Confirm with `java -version` and `echo $env:JAVA_HOME` in a **new** terminal after changes.

Step 4 — Install Maven 3.9.x and Git

Git: install from <https://git-scm.com/> if `git --version` fails.

Maven:

1. Download `apache-maven-3.9.x-bin.zip` from <https://maven.apache.org/download.cgi> or <https://archive.apache.org/dist/maven/maven-3/>
2. Extract to `C:\Program Files\Apache\maven\apache-maven-3.9.x`.
3. Create junction `C:\Program Files\Apache\maven\current` → that folder (optional but recommended).
4. System env: `MAVEN_HOME = C:\Program Files\Apache\maven\current`; add `%MAVEN_HOME%\bin` to Path.
5. New PowerShell:

```
mvn -version
git --version
```

Expected: Maven `3.9.x` reporting Java `21`.

Step 5 — Create `java-bootcamp` workspace

Where to run this: Open a **new Windows PowerShell** window from the Start menu (or Win + X → **Windows Terminal / Terminal**). Do **not** use IntelliJ’s terminal yet — that comes in Step 6 after you open the folder. You can start from any directory; the commands create `%USERPROFILE%\java-bootcamp` under your user home (for example `C:\Users\<You>\java-bootcamp`).

This layout matches what you use in every later lab: hands-on code under `examples\`, Lab 0 evidence under `notes\screenshots\`. Course lab handouts are provided separately by your instructor — you do **not** need a top-level `labs\` folder in this workspace.

```
$root = Join-Path $env:USERPROFILE 'java-bootcamp'
New-Item -ItemType Directory -Force -Path @(
    "$root\examples",
    "$root\notes\screenshots"
) | Out-Null
cd $root
Get-ChildItem
Get-ChildItem notes
```

Expected:

```
java-bootcamp\
  examples\      # HelloJava next; later labs: lab1-answers, labN-crm, ...
  notes\
    screenshots\ # Pass-criteria screenshots (redact secrets)
```

Step 6 — Open workspace in IntelliJ

1. **File** → **Open...** → %USERPROFILE%\java-bootcamp.
2. Trust the project if prompted.
3. **File** → **Project Structure** → **Project** → SDK **21**, language level **21**.
4. **View** → **Tool Windows** → **Terminal** → confirm `java -version`.

Optional VS Code: File → Open Folder... → same path; Terminal → New Terminal.

Step 7 — HelloWorld sources

```
cd $env:USERPROFILE\java-bootcamp
New-Item -ItemType Directory -Force -Path examples\HelloJava\src,
examples\HelloJava\out | Out-Null
```

Create `examples\HelloJava\src\HelloJava.java`:

```
public class HelloJava {
    public static void main(String[] args) {
        System.out.println("Hello Java Bootcamp!");
    }
}
```

Step 8 — Run from IntelliJ terminal

```
cd $env:USERPROFILE\java-bootcamp\examples\HelloJava
javac -d out src\HelloJava.java
```

```
java -cp out HelloJava
```

Expected:

```
Hello Java Bootcamp!
```

Step 9 — Run with IntelliJ green arrow

1. Open `HelloJava.java`.
2. Right-click `src` → **Mark Directory as** → **Sources Root** if needed.
3. Green ▶ beside `main` → **Run 'HelloJava.main()'**.

Step 10 — Git + identity (for later labs)

1. Create or sign in to a **GitHub** account (website: github.com).
2. Set Git to your **display name** and your GitHub **noreply email** (recommended — avoids push errors if your personal email is private on GitHub).

Find your noreply email: GitHub → **Settings** → **Emails** → enable **Keep my email addresses private** → copy the address shown, e.g. `{id}+{username}@users.noreply.github.com`.

```
git config --global user.name "Your Name"
git config --global user.email "12345678+yourusername@users.noreply.github.com"
git config --global --list
git --version
```

Expected: `user.name` and `user.email` appear in the list; `git --version` shows **2.x**.

If push fails with GH007 (“private email address”): your commit used a personal `@gmail.com` (or similar) address. Re-run Step 10 with the **noreply** email from GitHub Settings → Emails, or disable “Block command line pushes that expose my email” on that same page.

Keep personal verification screenshots under `%USERPROFILE%\java-bootcamp\notes\screenshots\` only — do not publish them.

Pass criteria (Windows)

```
java -version
javac -version
mvn -version
git --version
echo %env:JAVA_HOME
cd %env:USERPROFILE%\java-bootcamp
Get-Location
```

Mark each row **Pass** or **Fail** in your own notes.

#	Confirm	Your notes
1	Java / javac 21 (Temurin)	Pass / Fail
2	Maven 3.9.x on Java 21	Pass / Fail
3	Git works; <code>user.name</code> / <code>user.email</code> set	Pass / Fail
4	<code>JAVA_HOME</code> points at Temurin 21	Pass / Fail
5	Workspace <code>%USERPROFILE%\java-bootcamp</code> with <code>examples\</code> and <code>notes\screenshots\</code>	Pass / Fail
6	HelloWorld prints from terminal	Pass / Fail
7	HelloWorld runs via IntelliJ green arrow	Pass / Fail
8	(Optional) VS Code opens the same folder	Pass / Fail

Do not start Lab 1 until every Pass criteria row is Pass in your notes.

When complete, continue to **Lab 1** in your course materials.